American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
8	1/28/26	9 am	40 min	⦿ Advisor ○ Students	Lynn Ariss

Attendees:

Raghad Al Agha, Lynn Ariss, Hadi Elham, Claritta Khoury

Briefly summarize the main discussions during the meeting:

During the meeting, we discussed the integration of an additional sensor into the system, specifically the flex sensor, which will be used to represent the severity of edema through a bar indicator in the application interface. We also talked about improving the overall usability of the application, particularly considering that many users, especially elderly individuals, may not be familiar with medical terms such as edema. In addition, the team discussed the importance of supporting multiple languages in the application, since some users may prefer using Arabic or French instead of English. The meeting also included updates on technical progress, such as rebuilding parts of the pressure sensing circuit, performing soldering to improve hardware stability, eliminating cross talk in the pressure heatmap, integrating the pressure sensing data into the application, and working on making the BLE communication of the pressure matrix operate in real time. Finally, updates were shared regarding the conference submission, including writing the conference paper and the one-page abstract and successfully submitting them. The team also discussed updates regarding the PPG sensor. The current PPG signal was extracted prior to peak detection and its quality was verified by visualizing it in MATLAB. With the signal acquisition stage now complete, the remaining step is to correlate the extracted PPG signal with the data used for training.

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

The team concluded that the flex sensor will be integrated into the application and represented through a clear visual bar that indicates the level of edema, making the information easier to interpret. It was also agreed that the application interface should be made more user-friendly, with simplified terminology and clearer visual indicators to ensure accessibility for elderly users. Additionally, the team decided to move forward with integrating multiple language options, particularly Arabic and French, to improve accessibility for a wider range of users. On the technical side, the improvements made to the pressure sensing circuit and hardware stability will be maintained, while further work will continue on reducing cross talk in the pressure heatmap and ensuring that BLE communication for the pressure matrix operates in real time within the application. For the PPG sensor, the next step is to ensure that the PPG signal is processed in a format compatible with the machine learning algorithm, matching the structure of the data used during training.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Work on full system integration, work on charging/powering circuit, start preparing the device user manual	Raghad Al Agha	Feb 18
Optimize the mobile application interface and improve the overall design for better usability. Test and verify the integration of multiple languages in the app while continuing work on the model.	Lynn Ariss	Feb 18
Work on tuning pressure matrix velostat and integrate it in the app. Fix slow ble communication and work on Capsnet Ataxia Model	Hadi Elham	Feb 18
Work on full system integration, start preparing the device user manual, ensure PPG signal is processed in a format compatible with the ML algorithm	Claritta Khoury	Feb 18